MTH129 Introduction to AI for Mathematical Sciences (MS)

2025-2026 Academic Year
Semester 2

SMP AI Team

Xi'an Jiaotong-Liverpool University
西交利物浦大学

Logistic Regression

Table of Contents

1 Logistic Regression

2 Multiclass Extension: Softmax Regression

3 Logistic vs Linear

From Regression to Classification

Why not use linear regression for classification?

- Output $y \in \{0, 1\}$ discrete, linear regression gives continuous values
- Predictions outside [0,1] range are problematic
- Sensitive to outliers

Logistic Regression: A Probabilistic Approach

$$P(y = 1|\mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x}) = \frac{1}{1 + \exp(-\mathbf{w}^T\mathbf{x})}$$

- $\sigma(z)$: Sigmoid function (maps $\mathbb{R} \to [0, 1]$)
- Decision boundary: $\mathbf{w}^T\mathbf{x} = 0$

Likelihood Function for Logistic Regression

Probability model:

$$P(y_i|\mathbf{x}_i, \mathbf{w}) = \begin{cases} \sigma(\mathbf{w}^T\mathbf{x}_i) & \text{if } y_i = 1 \\ 1 - \sigma(\mathbf{w}^T\mathbf{x}_i) & \text{if } y_i = 0 \end{cases}$$

Compact form:

$$P(y_i|\mathbf{x}_i, \mathbf{w}) = \sigma(\mathbf{w}^T\mathbf{x}_i)^{y_i}[1 - \sigma(\mathbf{w}^T\mathbf{x}_i)]^{1-y_i}$$

Joint likelihood:

$$\mathcal{L}(\mathbf{w}) = \prod_{i=1}^{n} P(y_i|\mathbf{x}_i, \mathbf{w}) = \prod_{i=1}^{n} \sigma(\mathbf{w}^T\mathbf{x}_i)^{y_i}[1 - \sigma(\mathbf{w}^T\mathbf{x}_i)]^{1-y_i}$$

Log-likelihood

$$\ell(\mathbf{w}) = \sum_{i=1}^{n} [y_i \log \sigma_i + (1 - y_i) \log(1 - \sigma_i)]$$

where $\sigma_i = \sigma(\mathbf{w}^T\mathbf{x}_i)$

MLE Derivation: Gradient Calculation

Maximize log-likelihood $\ell(\mathbf{w})$, Key derivative: $\frac{d\sigma}{dz} = \sigma(1-\sigma)$

$$\begin{aligned}
\frac{\partial \ell}{\partial w_j} &= \sum_{i=1}^{n} \left[y_i \frac{1}{\sigma_i} \frac{\partial \sigma_i}{\partial w_j} - (1-y_i) \frac{1}{1-\sigma_i} \frac{\partial \sigma_i}{\partial w_j} \right] \\
&= \sum_{i=1}^{n} [y_i(1-\sigma_i)x_{ij} - (1-y_i)\sigma_i x_{ij}] \\
&= \sum_{i=1}^{n} [y_i x_{ij} - y_i \sigma_i x_{ij} - \sigma_i x_{ij} + y_i \sigma_i x_{ij}] \\
&= \sum_{i=1}^{n} (y_i - \sigma_i) x_{ij}
\end{aligned}$$

Gradient vector

$$\nabla_{\mathbf{w}} \ell(\mathbf{w}) = \sum_{i=1}^{n} (y_i - \sigma_i)\mathbf{x}_i = \mathbf{X}^T(\mathbf{y} - \boldsymbol{\sigma})$$

Optimization: No Closed-form Solution

Unlike linear regression:

- No analytical solution for $\nabla_{\mathbf{w}}\ell(\mathbf{w}) = 0$
- Log-likelihood is concave $\rightarrow$ unique global maximum

Gradient Ascent:

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} + \lambda \nabla_{\mathbf{w}}\ell(\mathbf{w}^{(t)})$$

Newton-Raphson Method (IRLS):

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \left(\nabla^2\ell(\mathbf{w}^{(t)})\right)^{-1} \nabla\ell(\mathbf{w}^{(t)})$$

$$\nabla^2\ell(\mathbf{w}) = -\mathbf{X}^T\mathbf{D}\mathbf{X}$$

$$\mathbf{D} = \text{diag}\{\sigma_i(1-\sigma_i)\}$$

Logistic Regression Numerical Example

Problem: Breast cancer diagnosis using two features

- Feature 1: Mean radius (normalized)
- Feature 2: Mean texture (normalized)
- Classes: Malignant (1), Benign (0)

Data and Initial Setup

$$\mathbf{X} = \begin{bmatrix} 1 & 1.0 & 0.5 \\ 1 & 0.8 & 0.7 \\ 1 & 0.3 & 0.2 \\ 1 & 0.1 & 0.1 \\ 1 & 0.5 & 0.3 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{w}^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \lambda = 0.1 \quad \text{(learning rate)}$$

Notation Clarification

- $\eta_i = \mathbf{w}^T \mathbf{x}_i$: Linear predictor (link function input)
- λ: Learning rate for gradient ascent

Step 1: Initial Probability Predictions

Compute linear predictor $\eta_i = \mathbf{w}^T\mathbf{x}_i$:

$$\eta_1 = 0 \cdot 1 + 0 \cdot 1.0 + 0 \cdot 0.5 = 0$$
$$\eta_2 = 0 \cdot 1 + 0 \cdot 0.8 + 0 \cdot 0.7 = 0$$
$$\eta_3 = 0 \cdot 1 + 0 \cdot 0.3 + 0 \cdot 0.2 = 0$$
$$\eta_4 = 0 \cdot 1 + 0 \cdot 0.1 + 0 \cdot 0.1 = 0$$
$$\eta_5 = 0 \cdot 1 + 0 \cdot 0.5 + 0 \cdot 0.3 = 0$$

Apply sigmoid function $\sigma(\eta) = \frac{1}{1+e^{-\eta}}$:

$$\sigma_1^{(0)} = \frac{1}{1+e^0} = 0.5 \quad \sigma_2^{(0)} = \frac{1}{1+e^0} = 0.5$$
$$\sigma_3^{(0)} = \frac{1}{1+e^0} = 0.5 \quad \sigma_4^{(0)} = \frac{1}{1+e^0} = 0.5$$
$$\sigma_5^{(0)} = \frac{1}{1+e^0} = 0.5$$

Step 2: Gradient Calculation

Gradient formula: $\nabla \ell = \mathbf{X}^T(\mathbf{y} - \boldsymbol{\sigma})$
Compute residual vector $\mathbf{r} = \mathbf{y} - \boldsymbol{\sigma}$**:**

$$\mathbf{r}^{(0)} = \begin{bmatrix} 1 - 0.5 \\ 1 - 0.5 \\ 0 - 0.5 \\ 0 - 0.5 \\ 0 - 0.5 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.5 \\ -0.5 \\ -0.5 \\ -0.5 \end{bmatrix}$$

Compute $\mathbf{X}^T\mathbf{r}$:

$$\mathbf{X}^T\mathbf{r} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1.0 & 0.8 & 0.3 & 0.1 & 0.5 \\ 0.5 & 0.7 & 0.2 & 0.1 & 0.3 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.5 \\ -0.5 \\ -0.5 \\ -0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \cdot 0.5 + 1 \cdot 0.5 + 1 \cdot (-0.5) + 1 \cdot (-0.5) + 1 \cdot (-0.5) \\ 1.0 \cdot 0.5 + 0.8 \cdot 0.5 + 0.3 \cdot (-0.5) + 0.1 \cdot (-0.5) + 0.5 \cdot (-0.5) \\ 0.5 \cdot 0.5 + 0.7 \cdot 0.5 + 0.2 \cdot (-0.5) + 0.1 \cdot (-0.5) + 0.3 \cdot (-0.5) \end{bmatrix}$$

$$= \begin{bmatrix} -0.5 \\ 0.45 \\ 0.3 \end{bmatrix}$$

Initial gradient:

$$\nabla \ell^{(0)} = \begin{bmatrix} -0.5 \\ 0.45 \\ 0.3 \end{bmatrix}$$

Step 3: Weight Update

Gradient ascent update: $\mathbf{w}^{(1)} = \mathbf{w}^{(0)} + \lambda \nabla \ell^{(0)}$

$$\mathbf{w}^{(1)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + 0.1 \begin{bmatrix} -0.5 \\ 0.45 \\ 0.3 \end{bmatrix} = \begin{bmatrix} -0.05 \\ 0.045 \\ 0.03 \end{bmatrix}$$

New weights:

$$w_0^{(1)} = -0.05, \quad w_1^{(1)} = 0.045, \quad w_2^{(1)} = 0.03$$

Step 4: First Iteration Results

New predictions with $\mathbf{w}^{(1)}$:

$$\eta_1^{(1)} = -0.05 + 0.045 \times 1.0 + 0.03 \times 0.5 = 0.01$$
$$\sigma_1^{(1)} = 1/(1 + e^{-0.01}) \approx 0.5025$$
$$\eta_2^{(1)} = -0.05 + 0.045 \times 0.8 + 0.03 \times 0.7 = 0.007$$
$$\sigma_2^{(1)} = 1/(1 + e^{-0.007}) \approx 0.5017$$
$$\eta_3^{(1)} = -0.05 + 0.045 \times 0.3 + 0.03 \times 0.2 = -0.0305$$
$$\sigma_3^{(1)} = 1/(1 + e^{0.0305}) \approx 0.4924$$
$$\eta_4^{(1)} = -0.05 + 0.045 \times 0.1 + 0.03 \times 0.1 = -0.0425$$
$$\sigma_4^{(1)} = 1/(1 + e^{0.0425}) \approx 0.4894$$
$$\eta_5^{(1)} = -0.05 + 0.045 \times 0.5 + 0.03 \times 0.3 = -0.0185$$
$$\sigma_5^{(1)} = 1/(1 + e^{0.0185}) \approx 0.4954$$

Log-likelihood improvement

Initial $\ell^{(0)} = -3.465$, New $\ell^{(1)} = -3.462$ (small increase)

Then, we can start the next iteration to improve the results further.

Step 5: Second Gradient Calculation

New residuals: $\mathbf{r}^{(1)} = \mathbf{y} - \boldsymbol{\sigma}^{(1)} = \begin{bmatrix} 1 - 0.5025 \\ 1 - 0.5017 \\ 0 - 0.4924 \\ 0 - 0.4894 \\ 0 - 0.4954 \end{bmatrix} = \begin{bmatrix} 0.4975 \\ 0.4983 \\ -0.4924 \\ -0.4894 \\ -0.4954 \end{bmatrix}$

Compute $\mathbf{X}^T\mathbf{r}^{(1)}$:

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1.0 & 0.8 & 0.3 & 0.1 & 0.5 \\ 0.5 & 0.7 & 0.2 & 0.1 & 0.3 \end{bmatrix} \begin{bmatrix} 0.4975 \\ 0.4983 \\ -0.4924 \\ -0.4894 \\ -0.4954 \end{bmatrix} = \begin{bmatrix} -0.481 \\ 0.441 \\ 0.291 \end{bmatrix}$$

New gradient:

$$\nabla \ell^{(1)} = \begin{bmatrix} -0.481 \\ 0.441 \\ 0.291 \end{bmatrix}$$

Step 6: Second Weight Update

Update weights: $\mathbf{w}^{(2)} = \mathbf{w}^{(1)} + \lambda \nabla \ell^{(1)}$

$$\mathbf{w}^{(2)} = \begin{bmatrix} -0.05 \\ 0.045 \\ 0.03 \end{bmatrix} + 0.1 \begin{bmatrix} -0.481 \\ 0.441 \\ 0.291 \end{bmatrix} = \begin{bmatrix} -0.0981 \\ 0.0891 \\ 0.0591 \end{bmatrix}$$

Updated weights:

$$w_0^{(2)} = -0.0981, \quad w_1^{(2)} = 0.0891, \quad w_2^{(2)} = 0.0591$$

Convergence and Final Model

After 5 iterations: $\mathbf{w}^* = \begin{bmatrix} -0.42 \\ 0.85 \\ 0.52 \end{bmatrix}$

Final predictions

Pt	P(Mal)	Pred/Actual	Correct
1	0.78	1/1	✓
2	0.76	1/1	✓
3	0.31	0/0	✓
4	0.29	0/0	✓
5	0.33	0/0	✓

Probability calculation: Pt1: $\sigma(-0.42 + 0.85 \cdot 1.0 + 0.52 \cdot 0.5) = 0.78$
Pt2: $\sigma(-0.42 + 0.85 \cdot 0.8 + 0.52 \cdot 0.7) = 0.76, \quad \cdots$

Accuracy: 100% (on training data)

Decision boundary:

$$-0.42 + 0.85x_1 + 0.52x_2 = 0$$

Key Observations from the Example

1. **Initialization:** Started from $\mathbf{w} = [0, 0, 0]^T$ (all probabilities 0.5)
2. **Gradient direction:**
   - Positive for features correlated with class 1
   - Negative for bias term (more class 0 samples)
3. **Convergence:** Weights stabilized after 5 iterations ($\lambda = 0.1$)
4. **Interpretation:**
   - $w_1 > 0$: Higher radius increases cancer probability
   - $w_2 > 0$: Higher texture increases cancer probability
   - $w_0 < 0$: Bias towards benign (more class 0)
5. **Model performance:** Perfect separation with linear boundary

Learning rate (λ) sensitivity

λ	Iterations	Final accuracy
0.01	50+	100%
0.1	5	100%
0.5	10	Diverged

Always standardize features for stable convergence!

Multiclass Extension: Softmax Regression

For K classes:

$$P(y = k|\mathbf{x}) = \frac{\exp(\mathbf{w}_k^T \mathbf{x})}{\sum_{j=1}^{K} \exp(\mathbf{w}_j^T \mathbf{x})}$$

Log-likelihood:

$$\ell(\mathbf{W}) = \sum_{i=1}^{n} \left[\mathbf{w}_{y_i}^T \mathbf{x}_i - \log \left(\sum_{k=1}^{K} \exp(\mathbf{w}_k^T \mathbf{x}_i) \right) \right]$$

Gradient

$$\nabla_{\mathbf{w}_k} \ell = \sum_{i=1}^{n} \left[\mathbb{I}(y_i = k) - P(y_i = k|\mathbf{x}_i) \right] \mathbf{x}_i$$

- Used in neural networks
- Generalization of logistic regression
- Cross-entropy loss

Logistic vs Linear: Key Differences

Aspect	Linear Regression	Logistic Regression
Output	Continuous	Probability (0-1)
Objective	Minimize SSE	Maximize log-likelihood
Solution	Closed-form $(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$	Iterative optimization
Error dist	Gaussian	Bernoulli
Decision boundary	Linear	Linear
Gradient	$\mathbf{X}^T(\mathbf{y} - \mathbf{X}\mathbf{w})$	$\mathbf{X}^T(\mathbf{y} - \boldsymbol{\sigma})$

When to use which?

- Linear: Predicting continuous outcomes (prices, quantities)
- Logistic: Binary classification (spam detection, diagnosis)
- Softmax: Multiclass classification (digit recognition)

Important Connection

Both are linear models: Logistic = Linear + Sigmoid Activation